Housing First or Treatment First?

A stylized model

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1 A stylized model

The quantitative model has six state variables and roughly forty parameters. That is what the welfare and fiscal questions require, but it makes it hard to see which piece of the environment does the work. This section strips the environment to the smallest one in which three questions are still well posed.

1. **Who uses drugs?** Use is concentrated among low income households. Which feature of the environment produces that, and can any other?
2. **What does housing assistance do to that decision?** A voucher pulls drug use in two directions at once. Which force is larger?
3. **When is Treatment First better than Housing First?** Where drug use harms nobody but the user, never. Addiction creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension alone cannot justify conditionality. The model's work here is to show why, and then to isolate and price the externality that can.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component has been removed and the model refitted without it, so what remains is what the results need.

2 The environment

2.1 Households and preferences

There are three periods and a continuum of households. A household is characterised by its permanent income y, fixed over its life and lognormally distributed, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma^2, \sigma^2)$, so mean income is one and every quantity below measured in units of income is a multiple of the population average. Households differ in y and in nothing else, so every household at a given income makes the same choice and drug use is a threshold in income rather than a probability. That also keeps intact the paper's claim to generate the income gradient without ex ante preference heterogeneity: there is none here to do the work.

Flow utility depends on non durable consumption c, the flow of housing services h consumed that period, and drug consumption v,

$$u(c, h, v) \;=\; \log c \;+\; \gamma \log h \;+\; \theta\big[\log(b+v) - \log b\big] \;+\; \aleph. \tag{1}$$

Here γ is the weight on housing services, which the household chooses each period from the three point grid $\{h_0, h_S, h_L\}$ of equation (5). The pair (θ, b) governs the drug term: marginal utility of the vth unit is $\theta/(b+v)$, so θ mostly sets how much a user uses and b whether anyone uses at all; the Stone and Geary form is what keeps the first unit's marginal utility finite, since under

plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it. It is what gives mortality a cost that does not vanish when consumption is low, and Section 4 turns on it. Households discount at β, save in one risk free asset at gross return R, and cannot borrow.

2.2 Timing

Period 1. Every household is employed and clean. It chooses consumption, housing, drug use $v \geq 0$ and saving, paying market rent on whatever unit it picks. Using generates the risk of a substance use disorder next period, rising in v,

$$\pi(v) \;=\; 1 - e^{-\lambda v}. \tag{2}$$

Because π is increasing, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

Period 2. For households that used, a disorder $d \in \{0, 1\}$ is realised with probability $\pi(v)$; for the rest it is not. The household is unemployed with probability u_0, or the higher u_1 if $d = 1$, and if unemployed receives income support. It then chooses consumption, housing and saving. The probability of recovering by the next period is ρ.

Period 3. Terminal. Employment, income support, the voucher and the housing choice work as in period 2; there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality. A household survives from one period to the next with probability

$$\phi(d) \;=\; \phi_0 \,(1 - \kappa d), \tag{3}$$

so κ is the proportional reduction in survival caused by the disorder.

2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the disorder: there is no wage penalty conditional on working, because the quantitative model sets that parameter to one. Income in a period is

$$y_e \;=\; y \;\text{ if employed,} \qquad y_e \;=\; \psi \cdot \min\{y, \bar{y}\} \;\text{ if unemployed,} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. The level of ψ links the disorder to homelessness: for the poorest households half of income falls short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income. Were income support above that rent nobody would lose a home and the moral hazard of Section 5 would be zero. The cap matters instead for Section 4: without it the resource cost of the disorder would be exactly proportional to income, and a penalty proportional to income cannot move a gradient that income itself generates.

Rent per unit of housing services p_r is the numeraire, so a unit's size and its market rent are the same number. Each period the household chooses

$$h \in \{h_0,\; h_S,\; h_L\}, \qquad h_0 < h_S < h_L, \tag{5}$$

a standard unit, a small unit, and homelessness, which is free and delivers almost nothing. Two rented sizes is the fewest that lets a household economise on housing rather than lose it; with one, homelessness comes out at 3.29 per cent against a target of 0.40 and no value of h_0 brings it back.

A voucher covers part of the rent, is means tested and is rationed. Writing $z \in \{0, 1\}$ for whether one is held,

$$\Pr(z = 1 \mid y_e, d) \;=\; \pi_V \cdot \mathbf{1}\{y_e \le y_V\} \cdot \mathbf{1}\{\text{policy covers } d\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option j is $r_j(z) = p_r h_j[\kappa_r z + (1 - z)]$, and zero when homeless. Vouchers operate in periods 2 and 3. The parameter π_V is the generosity of the programme and is the policy dial of Section 5.

The two policies differ in exactly one respect, the indicator in equation (6). **Housing First** covers every eligible household regardless of d. **Treatment First** covers only households without the disorder. Since cover in the disorder state is the only difference, the gap between them is the moral hazard and the whole of it.

2.4 The household's problem

Solving backward, with a assets carried in, e employment, z the voucher and d the disorder. In period 3 the household consumes everything and chooses housing; taking expectations over employment and the voucher lottery, both of which depend on d, gives the value of entering period 3,

$$W_3(a, d) = \mathbb{E}\Big[\max_j \big\{ \log(Ra + y_e - r_j(z)) + \gamma \log h_j + \aleph \big\} \,\Big|\, d \Big]. \tag{7}$$

In period 2 it chooses housing and saving, and faces mortality and recovery,

$$W_2(a, d) = \mathbb{E}\Big[\max_{j,\, a' \ge 0} \big\{ \log(Ra + y_e - r_j(z) - a') + \gamma \log h_j + \aleph + \beta\, \phi(d)\, \mathbb{E}_{d'} W_3(a', d') \big\} \,\Big|\, d \Big], \tag{8}$$

with $d' = 0$ with probability ρ if $d = 1$ and $d' = d$ otherwise. In period 1 it chooses housing, drug use and saving,

$$V_1(y) = \max_{j,\, v \ge 0,\, a' \ge 0} \Big\{ \log(y - r_j(0) - p_v v - a') + \gamma \log h_j + \theta[\log(b + v) - \log b] + \aleph + \beta\, \phi_0\, \mathbb{E}_{d'} W_2(a', d') \Big\}, \tag{9}$$

where the disorder arrives with the use dependent probability of equation (2). A household uses if the value at its best positive v exceeds the value at $v = 0$; because households differ only in y, that defines a set of income intervals on which use occurs.

3 Calibration

Table 1 lists the parameters set outside the model and their sources; Table 2 the three set inside it, with the moment each is matched to. The three periods are stages of a life rather than calendar intervals, and the parameters take conventional values without a strict mapping to a number of years. Nothing below turns on that: every comparison varies one object and holds the rest of the parameter vector fixed.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor	Standard	0.96
R	Gross interest rate	Standard	1.03
σ	Dispersion of log permanent income	Heathcote, Perri and Violante (2010)	0.75
ϕ_0	Survival per period, no disorder	CDC life tables	0.93
$\aleph$	Flow value of being alive	Viscusi and Aldy (2003)	2.50
γ	Weight on housing services	Consumer Expenditure Survey	0.25
θ	Weight on drug use	Set; b is calibrated given it	0.11
The disorder			
ρ	Recovery probability per period	SAMHSA, TEDS	0.40
κ	Proportional fall in survival with the disorder	Hjorthøj et al. (2023)	0.45
u_0	Unemployment probability, no disorder	BLS	0.049
u_1	Unemployment probability, with the disorder	NSDUH 2024, ages 26 and over, weighted	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
The externality on children			
δ_d	Income penalty to a child whose parent acquires the disorder	Transmission ratio 2.4; McGovern et al. (2023)	0.30
δ_h	Income penalty to a child whose parent is homeless	Chetty, Hendren and Katz (2016)	0.31
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Numeraire	1.00
p_v	Price of a unit of drugs	Normalisation of the units of v	1.00
h_L	Standard rental unit	ACS 2023, rent over income	0.212
h_S	Small rental unit	Half the standard unit	0.106
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD, share of eligible assisted	0.27

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00394
h_0	Housing services when homeless	Homelessness rate	0.399%	0.356%	0.00013
λ	Addiction hazard per unit of use	Share of users who acquire the disorder	33.2%	32.8%	97.1

Note: one untargeted moment is worth recording. The share of the homeless carrying the disorder comes out at 0.56, the right order but below the data.

4 Who uses drugs

Start from the force any explanation has to overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a disorder that is the only force operating, and drug use concentrates at the top: the average user earns 3.41 times the population average. Two features of the environment could reverse that. The *mortality channel* is that the disorder kills you, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home.

Figure 1 scales each from nothing to its calibrated value, recalibrating b at every point so participation stays at 5.13 per cent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

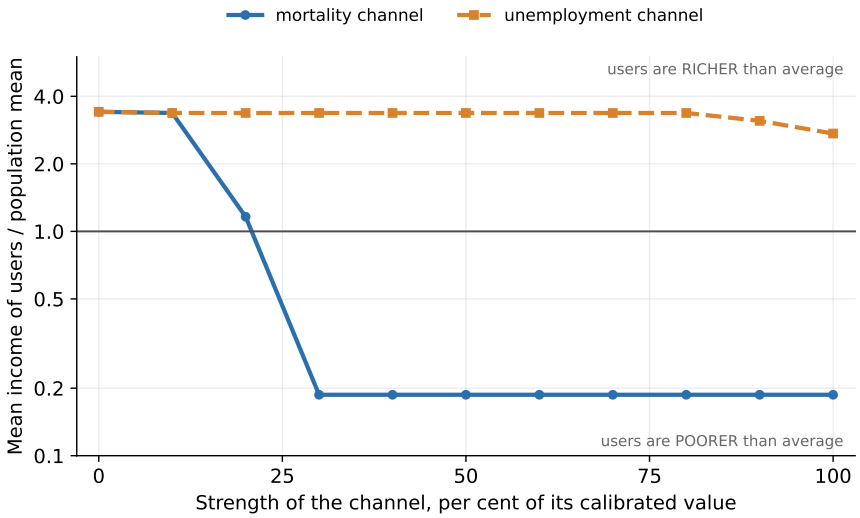


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over average income in the population, on a log scale. At every point b is recalibrated so participation is 5.13 per cent, so the curves compare economies with the same amount of drug use.

The result is sharper than the statement that mortality matters more. The mortality channel reverses the gradient at a quarter of its calibrated strength, taking the income ratio from 3.41 to 0.19, and does not move again. The unemployment channel does not reverse it at all, leaving users at 2.73 times average income even fully switched on. A resource penalty *cannot* produce the concentration of drug use at the bottom, however strong it is made.

The asymmetry is not obvious, and the same logic operates in the quantitative model. A resource penalty scales with the very thing it is meant to offset: taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. Indeed were income support exactly proportional to income the unemployment curve would be flat, and the fall to 2.73 is entirely the work of the cap $\bar{y}$ breaking that proportionality at the top. A mortality penalty is different in kind, because what it destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. This is why $\aleph$ appears in equation (1): it is the part of the continuation value that does not

shrink as consumption falls.

The mortality channel carries the policy result of Section 5 as well as this one. Switch it off and the deterrent effect of a voucher falls from -6.45 percentage points to -0.23, which is to say it disappears: a voucher deters nobody once what drug use threatens is only resources, because resources are what the voucher supplies either way.

5 What housing assistance does

A voucher acts on the decision to use through two opposing forces. It raises the value of remaining clean, since a housed future is worth more than a homeless one and using puts that future at risk, raising $W_2(a', 0)$ in equation (9): call this the value of life force, and it lowers use. It also cushions the consequence of acquiring the disorder, since a household that acquires it is housed anyway, raising $W_2(a', 1)$: call this the moral hazard force, and it raises use. Any real programme moves both at once, and the two policies separate them exactly, because under Treatment First the voucher is withdrawn on acquiring the disorder so only the first operates.

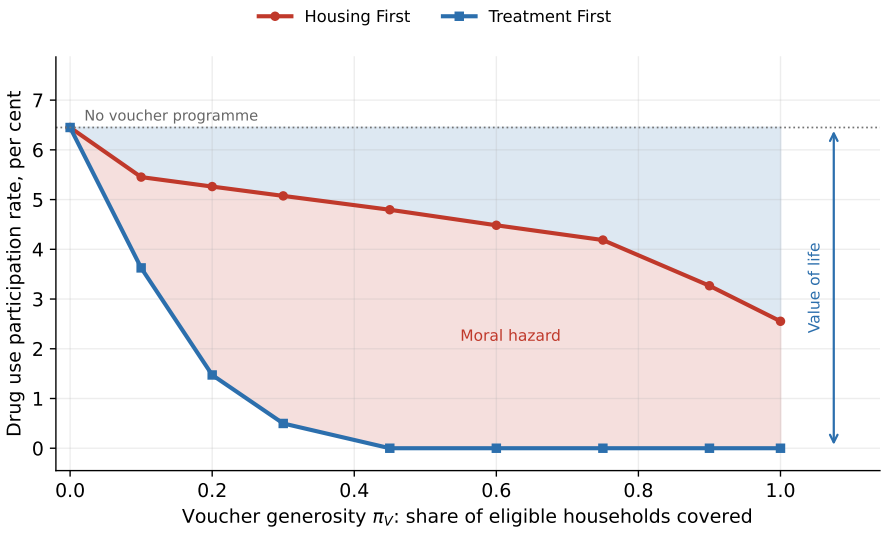


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard, the extra drug use Housing First produces relative to Treatment First at the same generosity; the blue band above it is what Housing First delivers net. Together they are the value of life force, marked by the arrow at the right.

Figure 2 traces both as generosity rises. With no programme, 6.45 per cent of households use. Universal Treatment First takes that to nothing, a value of life force of -6.45 percentage points; universal Housing First leaves it at 2.55 per cent, so the moral hazard is $+2.55$ points and the net effect of Housing First is -3.90. Both forces are first order and they work against each other, which is the central feature of the problem. Theory does not sign the effect of housing assistance on drug use, and anyone asserting that housing the homeless must increase it, or must reduce it, is making a quantitative claim while sounding as though they are making a theoretical one.

Drug use is not what a housing programme is for, and a reader shown only Figure 2 could conclude Treatment First wins on every count. Figure 3 plots the outcomes the programme is aimed at. On homelessness the two end in the same place, both driving it from 0.62 per cent to

zero, with a gap along the way never exceeding a tenth of a point: Housing First does not fail at housing people. The difference is entirely on the other margin, where it leaves 0.25 per cent of the population with the disorder and Treatment First leaves none. The moral hazard shows up as disorder cases among households who are housed, not as families left on the street.

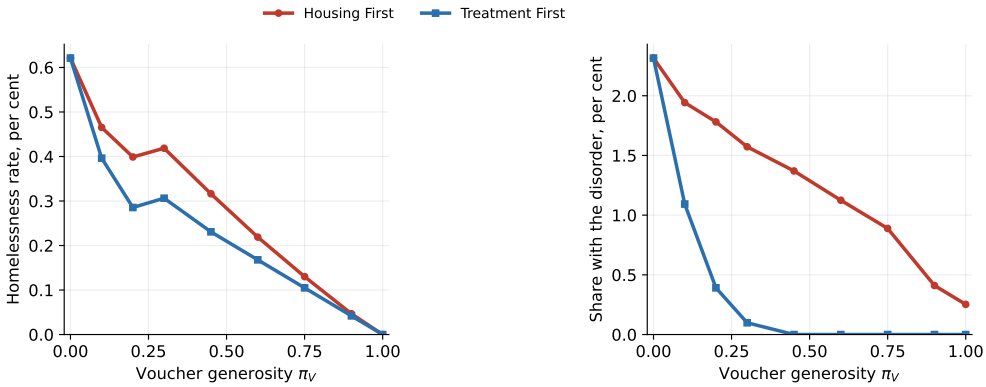


Figure 3: Homelessness and the disorder under the two policies

6 Welfare

6.1 Without externalities, Housing First always wins

Compare the policies at the same π_V. They differ only in the indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The household's budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly, by not using. Conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.

It follows that the moral hazard is not a distortion. Households here face the true addiction hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household taking a risk it is now better able to bear, as an insured household drives in worse weather. The +2.55 points of Section 5 is a statement about drug use and, on its own, none about welfare.

A fixed budget does not rescue conditionality either. Treatment First is slightly *more* expensive at universal cover, 0.1792 against 0.1791, so matching on cost rather than on generosity changes nothing. The reason is unattractive and true: a household in the disorder state draws only 79 per cent of the outlay a clean one draws, because survival to period 3 falls from 0.93 to 0.51, so the households the moral hazard adds to the rolls die sooner and stop costing money.

So a case for conditionality must come from outside. Households make no mistakes, so there is no internality; the policies cost the same, so there is no fiscal argument; and drug use here touches nobody but the user. The last is the one that can be measured rather than assumed, and Table 3

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	2.55%	0.00%
Share with the disorder	0.253%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.1791	0.1792
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		3.6%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.30%
Consumption equivalent loss, whole population		0.011%
Panel C. How large an externality would reverse this		
Disorder cases avoided by conditionality		0.253 pp
Critical social cost per case		0.12

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is the expected present value of spending on a household from period 2 on, as a multiple of average income.

says in advance how large it must be. Conditionality prevents 0.253 percentage points of disorder cases at a private welfare cost of 0.011 per cent of consumption, so a social cost of 0.12 of one household's lifetime income per case would be exactly enough to justify it.

6.2 The externality on children reverses it

The link has to run through income rather than through preferences. Letting the child of a user inherit a stronger taste, a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. Letting it inherit a greater susceptibility, a higher λ, fails the other way, since a child facing a riskier drug uses *less*. Income works on both counts, and it works because of Section 4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use. The income gradient does the whole job, which means the model transmits addiction across generations *without transmitting any preference for drugs*. A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the disorder}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{11}$$

which the parent does not internalise. General income persistence is absent because it is not an externality: it operates whether or not the parent uses.

The two penalties are set to different things. Panel A of Table 4 shows $\delta_d = 0.30$ makes a child of a parent who acquired the disorder 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four. For δ_h, Chetty, Hendren and Katz (2016) find children moved to a low poverty neighbourhood before age 13 earned 31 per cent more as adults, which is conservative here since growing up homeless is presumably worse. The two also do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's disorder rate	Ratio	Loss to the child
0.00	1.98%	1.00	0.00
0.20	3.56%	1.80	0.91
0.30 (calibrated)	4.82%	2.43	1.61
0.40	6.56%	3.31	2.61
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.00029
Disorder cases prevented			0.253 pp
Damage per case, one child			1.43
Externality gain from conditionality			0.00360
Gain over loss			12.3

Note: Panel A is computed at the baseline programme, $\pi_V = 0.27$, because the transmission ratio in the data comes from an economy with a rationed programme. Loss to the child is its consumption equivalent welfare loss as a multiple of one lifetime income. Panel B compares the policies at universal cover, with one child per household and no discounting between generations.

Panel B puts the two sides together and they are not close. The private welfare cost of conditionality is 0.00029 of a lifetime income, small because only 3.6 per cent of households are affected and each loses 0.30 per cent of consumption. Against that, conditionality prevents 0.253 percentage points of disorder cases, and each case costs one child 1.43 lifetime incomes. The externality gain is twelve times the private loss, and seven times at $\delta_d = 0.20$. Measured against the threshold of 0.12 set out above, the comparison is not close either. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

6.3 What this settles and what it does not

One feature of the model inflates that margin, and it is the right note on which to hand the question over. Treatment First drives drug use to *exactly* zero here, an artifact of having no preference heterogeneity: every household at a given income makes the same choice, so a policy strong enough to deter the marginal user deters every user. Nobody here faces the condition and fails to meet it, and the households who cannot comply are the entire empirical case for Housing First. This model exists to make the mechanisms visible, and it has. Establishing which way each comparison actually goes, with households who cannot get clean, a government budget, an equilibrium drug price and a proper life cycle, is what the quantitative model is for.

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SAMHSA (NSDUH 2024, TEDS); HUD (Point in Time counts, Housing Choice Voucher rules); BLS; ACS.